# Multi-Form Navigation and Picture Display

# What the program does

A small log-in program. Each division sees only its own pictures.

## 1

### Pick a division

The list box shows Design, Manufacturing and Sales.

## 2

### Type the code

Each division has its own password.

## 3

### See the pictures

A new form opens. It shows the first picture of that division.

## 4

### Next and Back

Next shows the next picture. Back returns to the log-in form.

# The parts of the project

Four forms, one module, two text files and nine pictures.

| Part | Type | Job |
| --- | --- | --- |
| Form1 | Form | Log-in: a list box, a password box and the CheckIn button |
| Form2 | Form | Shows the Design pictures |
| Form3 | Form | Shows the Manufacturing pictures |
| Form4 | Form | Shows the Sales pictures |
| Module1 | Module | Holds the variables that all forms share |
| DepartmentList.txt | Text file | The three division names |
| Image.txt | Text file | The nine picture file names |

# One picture list, three divisions

The program reads all nine names into one array. Each division owns three of them.

## Design

**Starts at index 0**

| | |
|---|---|
| **0** | Design1.jpeg |
| **1** | Design2.jpg |
| **2** | Design3.png |

## Manufacturing

**Starts at index 3**

| | |
|---|---|
| **3** | Mfg1.jpeg |
| **4** | Mfg2.jpeg |
| **5** | Mfg3.jpg |

## Sales

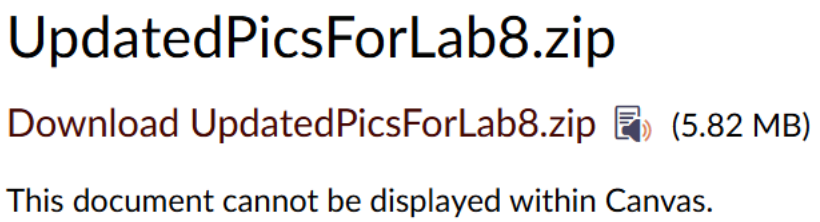
**Starts at index 6**

| | |
|---|---|
| **6** | Sales1.jpg |
| **7** | Sales2.jpg |
| **8** | Sales3.jpg |

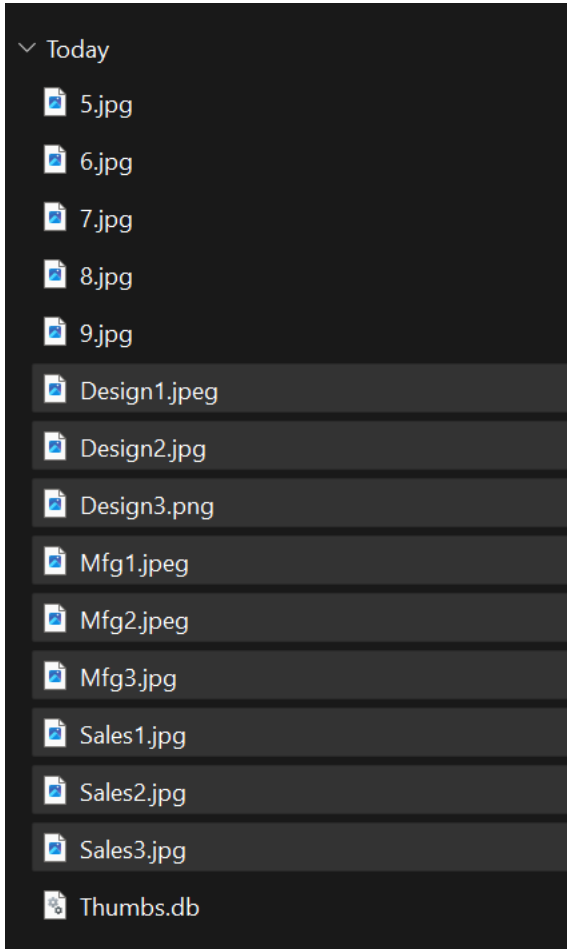
Form1 saves the start index in **target_Index**. The Next button goes from the start index to start + 2, then back to the start.

# Download the pictures from Canvas

You need nine pictures. Three for each division.




1. Open Canvas and go to Canvas\modules.
2. Download the file UpdatedPicsForLab8.zip.
3. Right-click the zip file and choose Extract All.
4. Find the nine pictures in the list on the right.



# Put the files in bin\Debug

The program looks for its files in bin\Debug, next to the .exe file.

1. Create a new project: Windows Forms App (.NET Framework), Visual Basic.
2. Press F5 one time, then close the program. This creates bin\Debug.
3. Copy the nine pictures into YourProject\bin\Debug.
4. In the same folder, create these two text files in Notepad.

**DepartmentList.txt**

```
"Design","Manufacturing","Sales"
```

**Image.txt**

```
"Design1.jpeg","Design2.jpg","Design3.png"
"Mfg1.jpeg","Mfg2.jpeg","Mfg3.jpg"
"Sales1.jpg","Sales2.jpg","Sales3.jpg"
```

Use double quotes and commas. Keep this order: Design, Manufacturing, Sales.

# Step 1: Add Module1 for shared variables

Project menu › Add Module › name it Module1.vb. Then type this code.

```
Module Module1
    ' Public = every form can read and change these variables.
    ' A form loses its own variables when it closes. These ones stay.

    ' The index of the first picture for the chosen division.
    ' Form1 sets it. Form2, Form3 and Form4 read it.
    Public target_Index As Integer

    ' Two string arrays. The () means "array". They are empty for now.
    ' Form1 fills them from the two text files.
    Public DepartmentList(), Imagelist() As String
End Module
```

# Step 2: Design Form1, the log-in form

Drag five controls from the Toolbox. Set the properties in the Properties window.

| Control | Name | Property to set | Job |
|---|---|---|---|
| Label | Label1 | Text = Division | Tells the user what the list is |
| ListBox | ListBox1 | Nothing | Shows the three division names |
| Label | Label2 | Text = Code | Tells the user what to type |
| TextBox | TextBox1 | PasswordChar = * | Hides the code while the user types |
| Button | Button1 | Text = CheckIn | Checks the code and opens the next form |

Keep the default names. The code on the next slides uses them.

# Form1 code: read the department list

Double-click an empty part of Form1. Visual Studio creates Form1_Load for you.

```
Public Class Form1
    ' One password for each division. Private = only Form1 can see it.
    Private password() As String

    Private Sub Form1_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ' i counts departments, j counts pictures, n is for the For loop.
        Dim i, j, n As Integer
        i = 0
        j = 0

        ' Open the file as file number 1. Input = we only read it.
        ' The name has no folder, so VB looks in bin\Debug.
        FileOpen(1, "DepartmentList.txt", OpenMode.Input)
        ' EOF = end of file. Keep reading until nothing is left.
        Do Until EOF(1)
            ' Make the array one item bigger. Preserve keeps the old items.
            ReDim Preserve DepartmentList(0 To i)
            ' Read one name from the file into the new item.
            Input(1, DepartmentList(i))
            i = i + 1
        Loop
        ' Close the file when you finish, so other code can use it.
        FileClose(1)
```

# Form1 code: read the pictures, fill the list

Still inside Form1_Load. This is the second half of the same Sub.

```
    ' The same steps for the picture names. This time it is file number 2.
    FileOpen(2, "Image.txt", OpenMode.Input)
    Do Until EOF(2)
        ReDim Preserve Imagelist(0 To j)
        Input(2, Imagelist(j))
        j = j + 1
    Loop
    FileClose(2)

    ' Show every department name in the list box.
    ' i is the number of names, so the last index is i - 1.
    For n = 0 To i - 1
        ListBox1.Items.Add(DepartmentList(n))
    Next

    ' Make room for one password per department. Then set each one.
    ' password(0) is for the first name in the list box, and so on.
    ReDim Preserve password(0 To i - 1)
    password(0) = "123"
    password(1) = "456"
    password(2) = "789"
End Sub
```

# Form1 code: the CheckIn button

Double-click the CheckIn button. Visual Studio creates Button1_Click for you.

```
Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
    ' SelectedIndex is the row the user clicked: 0, 1 or 2.
    ' It is -1 when no row is selected. Then no Case runs.
    Select Case ListBox1.SelectedIndex
        Case 0
            ' Design. Compare the typed text with the Design password.
            If TextBox1.Text = password(0) Then
                ' The Design pictures start at index 0 in Imagelist.
                ' target_Index is in Module1, so Form2 can read it.
                target_Index = 0
                ' Hide, not Close. Form1 is the start form.
                ' Closing it would end the whole program.
                Me.Hide()
                Form2.Show()
            Else
                ' Wrong code. Tell the user and stay on Form1.
                MsgBox("password is incorrect")
            End If
```

# Form1 code: the other two divisions

The same pattern. Three things change: the password, the start index and the form.

```
        Case 1
            ' Manufacturing. Its pictures start at index 3.
            If TextBox1.Text = password(1) Then
                target_Index = 3
                Me.Hide()
                Form3.Show()
            Else
                MsgBox("password is incorrect")
            End If

        Case 2
            ' Sales. Its pictures start at index 6.
            If TextBox1.Text = password(2) Then
                target_Index = 6
                Me.Hide()
                Form4.Show()
            Else
                MsgBox("password is incorrect")
            End If

    End Select
  End Sub
End Class
```

# Step 3: Design Form2, the Design form

Project menu › Add Form (Windows Forms) › keep the name Form2.vb.

| Control | Name | Property to set | Job |
|---|---|---|---|
| Label | Label1 | Text = Design | Shows the division name |
| PictureBox | PictureBox1 | SizeMode = StretchImage | Shows the picture |
| Button | Button1 | Text = Back | Returns to the log-in form |
| Button | Button2 | Text = Next | Shows the next picture |

StretchImage makes every picture fit the box. Our pictures have different sizes.

# Form2 code: the first picture, and Back

Double-click the form for Form2_Load. Double-click the Back button for Button1_Click.

```
Public Class Form2
    ' The picture this form shows now. Each form keeps its own index.
    Private index As Integer

    ' Load runs one time, when the form opens.
    Private Sub Form2_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ' Start at the first picture of this division.
        ' Form1 saved that number in target_Index.
        index = target_Index
        ' Imagelist(index) is a file name. Load shows that file.
        PictureBox1.Load(Imagelist(index))
    End Sub

    ' The Back button.
    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        ' Close this form. Then show the hidden log-in form again.
        Me.Close()
        Form1.Show()
    End Sub
```

# Form2 code: the Next button

Double-click the Next button. For Design, the index goes 0 → 1 → 2 → 0.

```
    ' The Next button.
    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        ' Each division has 3 pictures: target_Index, +1 and +2.
        ' So target_Index + 2 is the last picture of this division.
        If index < target_Index + 2 Then
            ' Not the last picture yet. Go one step forward.
            index = index + 1
        Else
            ' This is the last picture. Go back to the first one.
            index = target_Index
        End If
        ' Show the new picture.
        PictureBox1.Load(Imagelist(index))
    End Sub
End Class
```

# Step 4: Form3, the Manufacturing form

Add Form3 with the same four controls. Set the label text to Manufacturing.

```
Public Class Form3
    Private index As Integer

    Private Sub Form3_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ' Manufacturing starts at index 3. Form1 saved it in target_Index.
        index = target_Index
        ' Another way to show a picture. It does the same job as Load.
        PictureBox1.Image = Image.FromFile(Imagelist(index))
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        ' Back: close Form3 and return to Form1.
        Me.Close()
        Form1.Show()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        ' Next: the index goes 3, 4, 5, then back to 3.
        If index < target_Index + 2 Then
            index = index + 1
        Else
            index = target_Index
        End If
        PictureBox1.Image = Image.FromFile(Imagelist(index))
    End Sub
End Class
```

# Step 5: Form4, the Sales form

Add Form4 with the same four controls. Set the label text to Sales.

```
Public Class Form4
    Private index As Integer

    Private Sub Form4_Load(sender As Object, e As EventArgs) Handles MyBase.Load
        ' Sales starts at index 6. Form1 saved it in target_Index.
        index = target_Index
        PictureBox1.Image = Image.FromFile(Imagelist(index))
    End Sub

    Private Sub Button1_Click(sender As Object, e As EventArgs) Handles Button1.Click
        ' Back: close Form4 and return to Form1.
        Me.Close()
        Form1.Show()
    End Sub

    Private Sub Button2_Click(sender As Object, e As EventArgs) Handles Button2.Click
        ' Next: the index goes 6, 7, 8, then back to 6.
        If index < target_Index + 2 Then
            index = index + 1
        Else
            index = target_Index
        End If
        PictureBox1.Image = Image.FromFile(Imagelist(index))
    End Sub
End Class
```

# Step 6: Run and test

Press F5. Try every row in this table.

| Try this | You should see |
|---|---|
| Design with code 123 | Design1. Next shows Design2, Design3, then Design1 again. |
| Manufacturing with code 456 | Mfg1. Next shows only the three Mfg pictures. |
| Sales with code 789 | Sales1. Next shows only the three Sales pictures. |
| A wrong code | The message "password is incorrect". |
| The Back button | The log-in form again. |
| A "file not found" error | Check that the files are in bin\Debug. Check the names in Image.txt. |